Behind Closed Words: Creating and Investigating the forePLay Annotated Dataset for Polish Erotic Discourse

Warning: This paper contains examples of sexually explicit material

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Abstract

The surge in online content has created an urgent demand for robust detection systems, especially in non-English contexts where current tools demonstrate significant limitations. We introduce forePLay, a novel Polish-language dataset for erotic content detection, comprising over 24,000 annotated sentences. The dataset features a multidimensional taxonomy that captures ambiguity, violence, and socially unacceptable behaviors. Our comprehensive evaluation demonstrates that specialized Polish language models achieve superior performance compared to multilingual alternatives, with transformer-based architectures showing particular strength in handling imbalanced categories. The dataset and accompanying analysis establish essential frameworks for developing linguistically-aware content moderation systems, while highlighting critical considerations for extending such capabilities to morphologically complex languages.[1]

1 Introduction

The rapid growth of user-generated content online has created an urgent need for effective tools to detect and moderate harmful and inappropriate material. Traditional methods, such as manual review by editors or flagging by users, often fall short due to the sheer volume of content published daily. As a result, there has been an increasing reliance on automated solutions powered by advanced language models and natural language processing techniques.

Securing large language models (LLMs) against the generation of harmful content is another critical concern (Mahomed et al., 2024; Wu et al., 2024). While traditional content moderation tools are useful for user-generated content, they are less effective as input/output guardrails for LLMs due to their inability to adapt to new policies and distinguish between user-generated and AI-generated content (Inan et al., 2023). As a result, there is a growing need for datasets and specialized tools that can be integrated with LLMs to prevent the generation of explicit or harmful material, while maintaining their functionality across different applications.

While much of the previous work has focused on detecting toxicity, abusiveness, offensive language, or hate speech (Pavlopoulos et al., 2020; Gehman et al., 2020; MacAvaney et al., 2019), these categories do not encompass the full spectrum of undesired content (Markov et al., 2023). Erotic material, in particular, poses a significant risk, especially to minors. Detecting such content is essential not only for creating a safer online environment but also for upholding ethical standards. Although the Digital Services Act (DSA) primarily addresses non-consensual pornography and age-verification requirements, many online platforms, particularly those targeting minors, enforce stricter moderation policies that extend beyond legal obligations. Some platforms may prohibit explicit erotica or any 18+ material, even when it does not violate the law. One application of this research is to support such platforms in enforcing their internal guidelines, which vary depending on the service and its target audience.

Recent advances in deep learning have improved erotic content detection, but progress is hindered by the limited availability of training data, particularly open datasets. Existing tools are primarily designed for English-language content, limiting their effectiveness for other languages. This linguistic bias highlights the need for language-specific datasets and models that can capture the subtle semantic variations inherent in the expression of erotic content in different languages. Moreover, many existing datasets rely on overly simplified binary classification schemes, which fail to capture

[1]Dataset link: https://github.com/ZILiAT-NASK/ForePLay

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the complexities of erotic content, further limiting detection systems' effectiveness in diverse cultural and linguistic contexts.

To address these gaps, we introduce forePLay, the first Polish language manually annotated dataset of erotic content. In contrast to prior efforts, we present a novel multidimensional taxonomy applied to a large Polish language corpus ($n = 24,768$), incorporating fine-grained annotations across ambiguity, violence, and social unacceptability dimensions, while ensuring representation of LGBT-specific content from both literary and web-based sources. The corpus is constructed from user-generated content sourced from online fiction repositories and professionally published Polish literary works, offering diverse linguistic and stylistic coverage. The article provides a detailed description of the annotation process and discusses the challenges encountered during annotation, in particular human label variation. A major contribution is also the comprehensive evaluation of erotic content detection models, examining specialized Polish transformer-based models (HerBERT (Mroczkowski et al., 2021) and Polish RoBERTa (Dadas, 2023)) and Polish-specific LLMs such as the PLLuM family[2] and Bielik (Ociepa et al., 2024). We also compare these models with state-of-the-art multilingual and general-purpose models, including both open-source (Mixtral (Jiang et al., 2024), Llama 3.1 (Dubey et al., 2024), and C4AI Command-R (CohereForAI, 2024)) and commercial solutions (GPT-4o (Hurst et al., 2024)).

2 Related Work

2.1 Existing Datasets

Prior work has established several benchmark datasets for analyzing textual content with erotic themes, though these exhibit notable limitations in scope and annotation granularity. The most comprehensive corpus, Triplex, comprises 27,000 literary works totaling 1.62 billion tokens extracted from Archive of Our Own (Achour, 2016), while subsequent contributions include the erotic-books[3] corpus ($n = 646$) and the BeaverTails dataset containing a class of *sexually explicit, adult content* and developed for alignment research (Ji et al., 2023). Recent approaches to content analysis have leveraged large language models, as demonstrated in the erotica-analysis dataset[4] ($n = 15,000$) utilizing GPT-3.5 for automated annotation. Various targeted datasets address binary classification of explicit content, including the Jigsaw corpus and its derivatives ($n = 5,100$) (cjadams et al., 2019), though these typically employ simplified taxonomies. While specialized collections such as the sexting corpus[5] ($n = 547$) examine specific discourse patterns, existing datasets predominantly focus on English-language content with binary classification schemes.

2.2 Erotic Content Detection

Existing methodological approaches to erotic content detection span a broad spectrum of techniques, though their effectiveness is often constrained by the limitations of available training data. Early work relied on classical machine learning approaches, employing Support Vector Machines (SVM) and Random Forest classifiers trained on social media content (Barrientos et al., 2020), while simpler approaches utilized Naive Bayes classification for web page filtering (Hu et al., 2007). More recent neural approaches have demonstrated superior performance, as evidenced by Hierarchical Attention Networks applied to historical Latin texts (n = 2,500 sentences), achieving significant improvements over token-based methods (Clerice, 2024). Contemporary research has increasingly leveraged large language models, as demonstrated by CENSORCHAT (Qiu et al., 2024), which employs knowledge distillation for monitoring dialogue systems. Advanced architectures combining semantic and statistical features through Convolutional Neural Networks (CNN) have shown promise (Wu and Hu, 2005). However, these approaches predominantly focus on English language content, limiting their applicability to other languages where linguistic nuances and cultural contexts play crucial roles in content interpretation.

2.3 Content Moderation Systems

One key application of erotic content detection datasets is content moderation, covering both user-generated and AI-generated material. Specialized systems like Llama Guard, a Llama2-7b safeguard model developed by Meta, use safety taxonomies

[2]For more information, see https://pllum.org.pl/ and https://huggingface.co/CYFRAGOVPL

[3]https://huggingface.co/datasets/AlekseyKorshuk/erotic-books

[4]https://huggingface.co/datasets/openerotica/erotica-analysis

[5]https://github.com/mathigatti/sexting-dataset

to categorize prompts, including sexual content (Inan et al., 2023). However, comparative studies indicate that general-purpose LLMs, such as GPT-4o (Hurst et al., 2024) and Gemini 1.5 Pro (Team et al., 2024), often outperform these systems by achieving a better balance between false positives and false negatives (AlDahoul et al., 2024).

Holistic approaches to content moderation, such as those proposed by Markov et al. (2023), combine careful taxonomy design, active learning, and lightweight transformer models. These methods excel in detecting rare harmful content, such as material involving minors. Similarly, the framework introduced by Wu et al. (2024) leverages conceptual features from LLM inference, achieving high accuracy for sexual content with minimal computational cost. However, their findings reveal that LLMs tend to self-censor compared to human-written text. Similar conclusions were drawn by Mahomed et al. (2024), whose evaluation of OpenAI's moderation systems highlights GPT models' fine-tuning to avoid generating sexual content.

These findings highlight the challenges of building scalable, linguistically inclusive moderation frameworks, especially for nuanced categories like erotic material. Most current tools rely on English-centric training and evaluation pipelines (Markov et al., 2023). To address cultural and linguistic nuances, particularly in morphologically complex languages like Polish, language-specific detection strategies are essential.

3 Data

3.1 Data Collection

We constructed a large-scale Polish language corpus ($n = 24{,}768$ sentences) through systematic sampling from two distinct sources: (i) user-generated content from online fiction repositories and (ii) Polish literary works, including translations of world literature. The data collection was carried out by an interdisciplinary team consisting of literary scholars with extensive expertise in contemporary Polish literature, sociologists, NLP specialists, and IT engineers. The corpus comprises demographically diverse material drawn from 905 unique text units and includes a substantial, manually curated representation of LGBTQ+ narratives. This representation was ensured by selecting queer-themed literary texts—primarily original Polish works, supplemented by a smaller number of translated pieces—based on scholarly research and expertise in Polish queer literature. Document-level context is preserved through unique file identifiers. While direct URLs are omitted due to the volatility of online sources, our sampling methodology ensures broad coverage across linguistic registers and social contexts.

Given the potential application of models in supporting moderation, we focused on internet language and amateur writing, which is predominantly anonymous. Consequently, a significant portion of the sentences (69%) originates from online stories. While the majority of these stories are erotic and non-professional, we included non-erotic stories in a 1:4 ratio to enhance diversity and reduce genre-specific biases. To filter erotic content, we relied on tags and category labels related to erotica provided by the online repositories. All non-professional stories were scraped from publicly available websites, see Appendix A for details.

To avoid overfitting to specific individual writing styles, we limited the dataset to a maximum of two stories per author. Additionally, incorporating professional literary works with varying degrees of erotic themes (31%) aimed to further diversify linguistic patterns. When selecting these texts, availability in digital form was also a key criterion. A total of 22 different literary texts by Polish and international authors were included. This number is significantly lower than the count of unique text units for non-professional content. However, the literary texts were considerably longer, and finding works that met the established criteria proved challenging. Considering the overall proportion of professional works in the corpus, this number should be sufficient.

3.2 Data Preprocessing

Our text segmentation pipeline employs the NLTK library (Bird and Loper, 2004) for sentence boundary detection, which exhibits robust performance despite the inherent variability in web-sourced content quality. To maximize ecological validity, we preserved the original linguistic characteristics of the source material, including non-standard language patterns and orthographic variations, thereby enabling downstream models to generalize effectively to real-world applications.

The corpus contained a total of 342,546 tokens, as counted using the NLTK tokenizer, with an average sentence length of 13.83 tokens (median 11.0, std 11.53). Detailed token statistics are presented in Table 1.

Subcorpus	Mean	Std	25%	50%	75%
Total	13.83	11.53	7.0	11.00	17.00
Non-professional	12.43	10.32	6.0	10.00	16.00
Professional	16.90	13.31	9.0	14.00	21.00

Table 1: Token Count Statistics

Inconsistencies in text segmentation and discrepancies in automatically detected sentences arose from the fact that non-professional writers often did not use proper punctuation, making it challenging for the NLTK library to accurately identify sentence boundaries.

4 Annotation Process

To assemble the annotation team, three female annotators were allocated from our internal organization. To ensure gender balance, three part-time male annotators were recruited as external contractors. All selected annotators underwent training and guidance to maintain annotation consistency and quality throughout the project.

This process resulted in a gender-balanced team of six annotators (3 male, 3 female), aged 20–40. We included annotators with backgrounds in linguistics, literature, or related fields to ensure sensitivity to language nuances. To minimize potential bias, the annotators were not provided with metadata that could link specific samples to genre types. Furthermore, samples from each genre type were distributed evenly among all annotators. Each sentence was independently assessed by three annotators. Final labels were determined by majority vote. In cases where all three labels differed (830 out of $24,768$ samples, representing 3.35% of the total observations), a superannotator made the final decision. The superannotator, an NLP specialist and senior member of the annotation team, was responsible for overseeing consistency and quality throughout the annotation process. This role encompassed resolving disagreements among annotators, providing additional guidance, and conducting spot checks to ensure adherence to the defined guidelines.

4.1 Annotation Scheme

Based on the expert knowledge of the content, a set of five possible exclusive labels to annotate the samples was designed. This included the following categories: *erotic* (e), *ambiguous* (a), *violence-related* (v), *socially unacceptable behaviors* (u), and *neutral* (n). Some examples of the annotated text and its English translation are presented in Table 2.

Sentences labeled as *erotic* typically describe sexual activities and desires, advanced flirting with evident erotic undertones, references to past sexual activities, or explicit sexual fantasies. However, it is important to note that mere mention of genital terms or publicly accepted romantic behaviors, such as kissing or holding hands, does not warrant classification as sexual. Similarly, descriptions of physiological processes or neutral discussions of erotic or sexual topics (e.g., from the psychological perspective) are excluded from this category and considered neutral.

The *violence-related* category is reserved exclusively for sentences that include explicit sexual harassment, rape, lack of consent, or any other sort of non-consensual violence which concerns sexual intentions or activities. It is essential to distinguish consensual BDSM behaviors, as well as non-sexual acts of violence (based solely on the sentence-level analysis), as they do not fall under this category.

The category of *socially unacceptable behaviors* includes sentences that describe sexual behaviors considered illegal, generally taboo, or violative of social norms, such as zoophilia, necrophilia, pedophilia, incest, and other sexual deviations. This category takes precedence over *violence-related* label, meaning that if a sentence could be classified as both *socially unacceptable* and *violence-related*, it should be categorized under *socially unacceptable behaviors*.

Another label covers context-related sentences, which fall under the category of *ambiguous* samples. These sentences are identified based on the conviction that—given commonly recognized patterns of sexual descriptions—they evoke erotic or sexual connotations. However, in a neutral context, they could be interpreted as describing non-sexual behaviors or acts.

Lastly, sentences that did not fall into any of the above-mentioned categories were labeled as *neutral*. On one hand, these typically include non-sexual content on any topic. On the other, this category also comprises general statements about human sexuality or intimate parts that do not typically evoke erotic associations. While such sentences may be sex-related, they are not defined as erotic.

The annotation guidelines, developed by literary scholars and linguists based on domain expertise and material analysis, are described in detail in Appendix B.

Original sentence in Polish	English translation	Label
"Mój mężczyzna..."— usłyszałem szept i zanim się zorientowałem zostałem bez koszulki, a Ty siedziałaś naprzeciw mnie, całując zachłannie moje usta i błądząc dłońmi po torsie, lekko go drapiąc.	"My man..."— I heard a whisper and before I knew it I was left shirtless and you were sitting across from me, kissing my lips greedily and wandering your hands over my torso, lightly scratching it.	erotic
Był cały twardy.	He was all hard.	ambiguous
Ten gad ją zdradzał!	This rat was cheating on her!	neutral

Table 2: Examples of annotations in the dataset

Subcorpus	Label Categories				
	Erotic	Ambiguous	Violence	Unacceptable	Neutral
Total	6,361	1,344	69	116	16,878
Non-professional	2,937	1,277	52	95	12,649
Professional	3,424	67	17	21	4,229

Table 3: Distribution of Label Categories Across Dataset Subcorpora

Following the completion of the superannotation process, the final dataset consists of two primary classes: 68.14% of the samples are labeled as *neutral*, and 25.68% as *erotic*. The remaining categories are minor, with 5.43% labeled as *ambiguous*, 0.47% as *socially unacceptable behaviors*, and 0.28% as *violence-related*. Distribution of labels across the subcorpora are presented in Table 3.

The percentage of violence-related examples, as well as those representing socially unacceptable behaviors, was scarce, which was expected, as the data collection process did not specifically target texts exploring such themes. Additionally, while only 15% of the texts in the dataset were sourced from non-erotic stories, the randomization process employed to minimize bias resulted in the inclusion of neutral sentences even from texts classified as erotic, as these are not exclusive to non-erotic stories.

4.2 Annotation Quality

The process of quality evaluation revealed various challenges faced by the annotators, highlighting the demanding nature of the task. The designed labels were intended to be applied exclusively, meaning that each sample could only be categorized as either neutral, sex-related, or ambiguous. Within the sex-related category, overlaps were mitigated by a clear hierarchical structure. As previously mentioned, in cases where a sample exhibited characteristics of both violence and socially unacceptable behaviors, priority was given to the latter. All the other sex-related samples contributed to the erotic class. Due to the distinctiveness of rare cases of sexual violence and non-acceptable sexual behaviors, this aspect did not pose significant concerns. Annotators applied these two labels only incidentally, with individual frequencies ranging from 0.2% to 0.6%.

Firstly, the annotation task itself—particularly given the wide range of non-normative language patterns found in non-professional short stories—was inherently subjective. Secondly, the prevalence of misspellings, controversial vocabulary, and grammatical errors, which diminished the clarity of the authors' intentions and made rational decision-making exceedingly difficult. Thirdly, the category of erotic sentences turned out to be extremely broad, spanning from evident and explicit samples, often including vulgar language, to much subtler erotic tones. Consequently, the *ambiguous* label, designed to account for context-related uncertainty, offered a means of resolving indecision between *sexual* and *neutral* classifications. This created a potential risk of overusing the ambiguous label as a way to mitigate uncertainty in borderline cases.

Human label variation (HLV) (Plank, 2022), inherently present in our task of annotating erotic sentences, raised several questions regarding the dataset design (see the Section 7). To preserve the authenticity of the annotation process, we chose not to intervene beyond the initial training phase. For the dataset's initial release, we opted to follow the conventional practice of data aggregation featuring a ground truth.

To guarantee annotation reliability, we calculated Cohen's Kappa for each pair of annotators across all categories. Furthermore, we evaluated inter-annotator agreement for each category individually. Detailed results are presented in Table 4.

Agreement in most pairs was consistent, ranging from 0.66 to 0.71, regardless of annotators' gender. However, pairs involving annotator Fem1 presented a notable deviation, as Fem1 emerged as an outlier,

Annotators	all	a	e	u	v
Fem1 / M1	0.1750	0.0532	0.3602	0.2545	0.1984
Fem1 / Fem2	0.1808	0.0370	0.3193	0.3952	0.2911
Fem1 / Fem3	0.1415	0.0151	0.2890	0.2740	0.2271
M1 / Fem2	0.7044	0.3505	0.7513	0.3728	0.4602
Fem2 / M2	0.6910	0.0833	0.7149	0.5074	0.1696
Fem2 / M3	0.6610	0.0037	0.6923	0.3973	0.3237
Fem2 / Fem3	0.7187	0.2683	0.7728	0.6718	0.4078
M2 / M3	0.7030	0.0368	0.7261	0.4191	0.2481

Table 4: Cohen's Kappa values for each annotator pair across all categories and for each individual category, broken down by gender (Fem – female, M – male).

largely due to an excessive use of the ambiguous label.

In addition to pairwise agreement, the overall consistency of the entire sample was assessed using Krippendorff's Alpha. The value was quite low (0.387). However, after excluding the Fem1 annotation, which significantly deviated from the other annotators, the value increased to 0.716, indicating a satisfactory level of agreement in the annotation process.

5 Experiments

We utilize a diverse set of language models for erotic content detection, ranging from specialized Polish encoder-based models and developed Polish-specific LLMs to state-of-the-art commercial LLMs. This variety enables a comprehensive comparison across different model architectures and training methodologies. Additionally, the translated English prompt for the LLMs is included in the appendix on Figure 2.

5.1 Datasets

The original dataset consists of five distinct label categories, but due to significant class imbalance, additional experiments were conducted with modified label groupings as outlined below:

- **Basic**: A binary classification setup where the dataset was reduced to two classes: *neutral* and *erotic* labels.
- **Core**: A three-class classification setup where the dataset included *neutral*, *erotic*, and *ambiguous* labels.
- **Extended**: A four-class classification setup that expanded the dataset to include *neutral*, *erotic* and *ambiguous* labels, with a merged category combining *violence-related* and *socially unacceptable behaviors*.
- **Full**: The full dataset containing all five original categories.

5.2 Specialized Polish Transformer-based Models

Encoder-based models for Polish were selected based on the KLEJ Benchmark leaderboard (Rybak et al., 2020). Two top-performing models across multiple downstream tasks were chosen: HerBERT and Polish RoBERTa. HerBERT is a Polish-language adaptation of BERT, trained on a Polish corpus, while Polish RoBERTa is an optimized version featuring a unigram tokenizer, whole-word masking, and an expanded vocabulary of 128k entries. In our experiments, both models were fine-tuned on each dataset configuration (**Basic**, **Core**, **Extended**, **Full**).

5.3 Polish Large Language Models

Our evaluation framework encompasses three variants from the PLLuM (Polish Large Language Model) family: **Llama-3.1-8B-PLLuM**, **PLLuM-Mistral-12B**, and **PLLuM-Mixtral-8x7B**, adapted from their respective base architectures. These models underwent pre-training on 147B Polish language tokens and subsequent instruction tuning. We additionally employed **Bielik 2.3**, trained on 200B tokens and aligned using DPO-positive on 66,000 examples. To assess adaptation capabilities, we fine-tuned Llama-3.1-8B-PLLuM and PLLuM-Mistral-12B on our classification task. All base models were evaluated under 0-shot, 1-shot, and 5-shot configurations, while fine-tuned variants underwent standard supervised evaluation.

5.4 State of the Art Large Language Models

For comparison with LLMs fine-tuned strictly for Polish, we employed a selection of state-of-the-art multilingual and general-purpose models, including both open-sourced and commercial solutions. Among these, **Mixtral**, characterized by its modular and scalable architecture, was utilized in both the 8x22B and 8x7B configurations, alongside **Mistral 12B**, designed as a lightweight alternative. We also tested **Llama 3.1 70B-Instruct** and its smaller counterpart **Llama 3.1 8B-Instruct**, both fine-tuned for instruction-following tasks, as well as **C4AI Command-R**, developed for command-following applications. Additionally, we incorporated **GPT-4o**, frequently used for tasks requiring generalization across various domains.

6 Results

Table 5 highlights the macro-F1 performance metrics of the HerBERT and RoBERTa models across classification tasks involving different dataset configurations. In binary classification scenarios, both models exhibit robust performance. However, as the number of classes increases, a decline in macro-F1 scores is observed. Notably, RoBERTa, particularly in its Base configuration, demonstrates superior performance relative to HerBERT, with pronounced advantages evident in tasks on the **Basic** and **Extended** datasets.

Dataset	HerBERT		RoBERTa	
	Base	Large	Base	Large
Basic	0.929	0.939	**0.944**	0.943
Core	0.702	0.738	0.738	**0.748**
Extended	0.693	**0.746**	0.704	0.734
Full	0.632	0.648	**0.707**	0.664

Table 5: Macro-F1 scores for HerBERT and RoBERTa models across different dataset versions.

In the case of LLMs, models from the PLLuM family consistently demonstrate strong performance across varying label counts and shot configurations, with particular emphasis on the PLLuM-Mixtral-8x7B, which achieves outstanding macro-F1 scores across a wide range of label counts, notably reaching 0.921 for **Basic** data and 0.670 for **Core** data in the 5-shot setting. Additionally Llama-3.1-8B-PLLuM (SFT) and PLLuM-Mistral-12B (SFT) show significant improvements when subjected to supervised fine-tuning, resulting in enhanced performance, with the highest macro-F1 scores observed across various label counts.

Fine-tuned LLMs exhibit performance levels comparable to traditional transformer-based architectures when evaluated on datasets labeled as **Basic** and **Core**, where all categories are sufficiently represented. This suggests that in scenarios characterized by balanced class distributions, where each class is supported by an adequate number of training examples, the performance differential between these methodological approaches is minimal. Conversely, for datasets labeled as **Extended** and **Core**, which are marked by pronounced class imbalance, models built upon the BERT architecture begin to demonstrate a distinct advantage. These models excel in effectively addressing the challenges posed by imbalanced data distributions, exhibiting superior capability in identifying underrepresented categories. This highlights their enhanced adaptability and robustness in handling more complex scenarios involving skewed training data distributions.

In the context of multilingual models trained with a primary emphasis on the English language, the results are generally inferior compared to models specifically adapted for the Polish language. The performance varies depending on the configuration, including the number of labels and the number of shots, with different models excelling under different conditions. However, overall, the Mixtrals and GPT-4o models consistently yielded the most favorable results.

The full results of all evaluated models—including the baseline models, the ones adapted to the Polish language from the PLLuM family, and the ones further fine-tuned on our dataset—are presented in Table 6. A detailed account of refusals and responses that could not be classified into any of the predefined categories is presented in Table 7. Among the evaluated models, Llama 3.1 8B-Instruct recorded the highest number of refusals, reaching a total of 848. In contrast, models that consistently provided clear and categorizable outputs included GPT-4o and Mixtral 8x22B, as well as those additionally fine-tuned on our dataset.

7 Discussion

The task of multi-dimensional eroticity annotation is highly complex and inherently subjective, necessitating further and more advanced reflection on the Human-Language Variation (HLV) framework, which has become an increasingly important open problem in recent years (Plank, 2022; Uma et al., 2021; Pavlick and Kwiatkowski, 2019). The researchers highlight the fact that disagreements in the annotation process should not be dismissed as mere *noise* in the search for a single ground truth. On the contrary, such disagreements can often reflect plausible alternative judgments, suggesting that these annotations may not always be categorical (De Marneffe et al., 2012). However, as demonstrated by our experience evaluating annotation quality, distinguishing between clear errors—where no plausible argument could justify the choice of a particular label—and alternative judgments is a critical issue that warrants deeper concern and further discussion. Within the context of erotic content detection, we believe it is valuable to advance current research by utilizing

Model	Basic			Core			Extended			Full		
	0-shot	1-shot	5-shot	0-shot	1-shot	5-shot	0-shot	1-shot	5-shot	0-shot	1-shot	5-shot
GPT-4o	0.888	0.873	0.891	0.640	0.605	0.618	0.425	0.410	0.415	0.340	0.334	0.349
C4AI Command-R	0.684	0.730	0.714	0.420	0.486	0.454	0.322	0.374	0.363	0.257	0.292	0.299
Llama 3.1 8B-Instruct	0.789	0.736	0.709	0.562	0.517	0.514	0.388	0.364	0.378	0.325	0.283	0.316
Llama 3.1 70B-Instruct	0.837	0.845	0.846	0.595	0.641	0.630	0.396	0.417	0.408	0.329	0.344	0.364
Mistral 12B	0.872	0.847	0.862	0.58	0.579	0.610	0.423	0.411	0.42	0.354	0.349	0.338
Mixtral 8x7B	0.867	0.87	0.862	0.616	0.622	0.621	0.440	0.451	0.435	0.364	0.364	0.348
Mixtral 8x22B	0.872	0.854	0.833	0.633	0.642	0.610	0.424	0.440	0.406	0.335	0.361	0.344
Bielik-11B-v2.3-Instruct	0.848	0.858	0.868	0.601	0.597	0.607	0.427	0.416	0.423	0.312	0.437	0.48
PLLuM-Mistral-12B	0.894	0.842	0.844	0.656	0.610	0.613	0.456	0.441	0.427	0.401	0.359	0.401
PLLuM-Mixtral-8x7B	0.874	0.902	0.921	0.647	0.662	0.670	0.452	0.456	0.426	0.422	0.406	0.422
Llama-3.1-8B-PLLuM	0.641	0.737	0.709	0.425	0.446	0.425	0.301	0.324	0.326	0.314	0.274	0.273
PLLuM-Mistral-12B (SFT)		0.946			**0.792**			0.458			**0.488**	
Llama-3.1-8B-PLLuM (SFT)		**0.947**			0.764			**0.461**			0.435	

Table 6: Macro-F1 scores for various models across different dataset configurations and shot numbers.

Model	Basic			Core			Extended			Full			Total
	0-shot	1-shot	5-shot	0-shot	1-shot	5-shot	0-shot	1-shot	5-shot	0-shot	1-shot	5-shot	
GPT-4o	0	0	0	0	0	0	0	0	0	0	0	0	**0**
C4AI Command-R	9	2	14	9	5	10	7	1	9	4	1	14	95
Llama 3.1 8B-Instruct	42	150	12	38	99	14	134	62	8	66	208	15	848
Llama 3.1 70B-Instruct	8	0	0	33	0	0	46	0	0	49	0	0	136
Mistral 12B	0	0	0	0	0	0	0	0	1	0	0	0	1
Mixtral 8x7B	0	0	2	1	0	2	1	5	7	3	0	4	25
Mixtral 8x22B	0	0	0	0	0	0	0	0	0	0	0	0	**0**
Bielik-11B-v2.3-Instruct	116	1	0	115	2	0	148	1	3	2	5	1	394
PLLuM-Mistral-12B	3	5	1	3	2	3	8	7	0	6	10	5	53
PLLuM-Mixtral-8x7B	0	0	1	0	0	2	0	0	2	0	1	2	8
Llama-3.1-8B-PLLuM	1	4	0	1	4	3	1	3	5	0	4	1	27
PLLuM-Mistral-12B (SFT)		0			0			0			0		**0**
Llama-3.1-8B-PLLuM (SFT)		0			0			0			0		**0**

Table 7: Number of unidentified responses or refusals.

datasets with multiple plausible labels and exploring the approach of Learning with Disagreements (LeWiDi) (Uma et al., 2021) or Multiple Ground Truth (Anand et al., 2024). This approach could offer more nuanced insights into the field by comparing the performance of models trained on aggregated annotations versus those trained on multiple possible labels.

Our experimental results demonstrate clear performance patterns across model architectures and linguistic contexts, with specialized Polish language models consistently outperforming multilingual alternatives in erotic content detection. The encoder-based architectures, particularly RoBERTa, achieve robust macro-F1 scores in binary classification ($0.929 - 0.944$), though performance degrades with increasing categorical complexity (dropping to ~ 0.66 for five-class classification). This degradation pattern reveals fundamental challenges in maintaining discriminative power across finer-grained categorical distinctions, particularly in cases requiring subtle cultural and contextual understanding.

The PLLuM family of models exhibits exceptional few-shot learning capabilities, with PLLuM-Mixtral-8x7B achieving macro-F1 scores of 0.921 in 5-shot settings for binary classification, while supervised fine-tuning of PLLuM-Mistral-12B yields the highest overall performance (macro-F1 = 0.946). However, the diminishing returns observed with increasing model size, coupled with the persistent challenge of handling ambiguous content in imbalanced datasets, suggests that architectural sophistication alone may not overcome the fundamental challenges of erotic content detection in morphologically complex languages. These

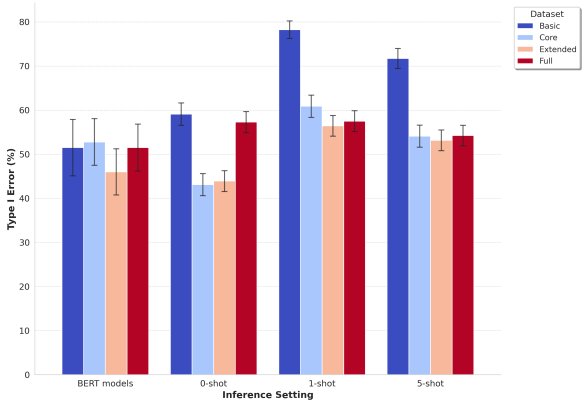


Figure 1: Type I errors percentage across datasets and inference settings. Error bars represent 95% confidence interval.

findings underscore the importance of language-specific model development for sensitive content detection tasks, rather than relying solely on increasingly large multilingual models.

Beyond overall performance metrics, understanding error patterns is crucial for content moderation deployment, where both false positives (over-censorship) and false negatives (missed inappropriate content) carry significant consequences. We calculated Type I error percentages as the number of false positives divided by the sum of false positives and false negatives to assess model precision across different classification scenarios. Full results regarding Type I and Type II errors for each model are presented in the Appendix D. The analysis reveals that Type I error rates decrease substantially with classification complexity—from approximately 60-80% in binary classification (Basic) to 40-50% in the full five-class scenario (Figure 1). This pattern indicates higher Type I error rates in simpler tasks and higher Type II rates in complex multi-class scenarios, highlighting the fundamental trade-off between precision and sensitivity in erotic content detection tasks.

8 Conclusions

This work presents forePLay, the first comprehensive Polish language dataset for erotic content detection, introducing a novel multidimensional taxonomy encompassing ambiguity, violence, and social acceptability dimensions. Through systematic empirical evaluation across multiple model architectures, our analysis demonstrates the superiority of specialized Polish language models over multilingual alternatives, particularly in handling nuanced categorical distinctions. The performance patterns observed in transformer-based architectures and few-shot learning scenarios provide valuable insights for developing language-specific content moderation systems. Our findings underscore the critical importance of culturally and linguistically adapted approaches in content detection tasks, while the presented dataset and evaluation framework establish essential groundwork for future research in multilingual content moderation systems, particularly for morphologically complex languages.

9 Data Availability

A subset of erotic and ambiguous sentences, totaling 3,704 samples, has been released on our public Github account. Due to ethical considerations, we opted to exclude the minor harmful classes of *violence-related* content and *socially unacceptable behaviors*. ForePLay Dataset Release 1.0 consists of 2,728 sentences labeled as *erotic*, and 976 sentences labeled as *ambiguous*, which corresponds to 43% and 73% of the original dataset, respectively. Neutral sentences were not included in this release, as such data is readily obtainable from publicly available corpora and may be easily used for comparative purposes in downstream tasks. This subset has undergone additional copyright verification and was made available following legal consultations. The released data and its license comply with new legal regulations that came into effect after the data collection and annotation process had been completed.

Limitations

Our study reveals several important limitations that warrant discussion. While our dataset represents a significant contribution to Polish language resources, it exhibits sampling biases through its predominance of content from online fiction repositories (69%). This may not fully capture the linguistic diversity of erotic discourse across different domains and registers. The relative underrepresentation of professional literary works (31%) potentially limits our understanding of more sophisticated expressions of erotic content.

A significant methodological limitation emerges from our binary approach to violence and socially unacceptable behaviors. The current taxonomy, while practical for annotation purposes, may oversimplify the complex spectrum of content severity

and social acceptability. This simplification could impact the generalizability of our models to real-world content moderation scenarios where more nuanced distinctions are crucial.

The annotation process presents limitations stemming from inherent subjectivity in erotic content interpretation. Despite achieving reasonable inter-annotator agreement (Krippendorff's Alpha of 0.716 after outlier removal), the significant variation in individual annotator interpretations—particularly evident in the use of the *ambiguous* category—suggests potential instability in ground truth labels.

From a technical perspective, our evaluation framework primarily focuses on classification accuracy without deeply examining model behavior on edge cases or adversarial examples. While our study demonstrates the superiority of Polish-specific models over multilingual alternatives, we acknowledge limitations in comparing across model architectures with significantly different parameter counts and training regimes.

Furthermore, a specific limitation pertaining to the evaluation phase is that the definitions of the labels were not explicitly included in the prompts provided to the models. We confirm this omission may have contributed to the lower performance of the models in comparison to the baseline.

These limitations underscore the need for future work addressing dataset diversity, annotation methodology refinement, and more robust evaluation frameworks for erotic content detection systems.

Acknowledgements

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A Data Sources

All non-professional stories were scraped from publicly available websites including:

- *opowiadaniaerotyczne-darmowo.com*
- *sexopowiadania.pl*
- *pornzone.com*
- *anonserek.pl*
- *opowi.pl* (categories: *o życiu, różne, miłosne*)
- *opowiadania.pl*
- *polki.pl*

B Annotation Guidelines Summary

This annotation task aims to create a manually annotated dataset for erotic content detection, enabling the development of automated detection methods and content moderation systems. These systems can assist models in identifying undesired content that may be potentially harmful. Each annotator is provided with a set of samples to label exclusively with one of the five predefined categories. The annotation process is conducted independently, meaning that annotators do not have access to the labels or work of other team members. For the purpose of this task, *a sample* is defined as a single sentence of varying length. The style and spelling of the sentences may vary significantly, and assessing the quality or correctness of the text is not part of the task.

Annotators are instructed to carefully read each sentence and assign one final label to it. Each sentence was presented in isolation, without additional context. They are also encouraged to pause or discontinue the task at any point if they feel uncomfortable with the content being evaluated.

Each sample should be exclusively assigned one of the following labels:

- Label *e* (for *erotic*): choose this label if the sentence describes sexual activities and desires, advanced flirting with evident erotic undertones, references to past sexual activities, or explicit sexual fantasies.
  Please note that mere mentions of genital terms or publicly accepted romantic behaviors or neutral discussions of erotic or sexual topics (e.g., from the psychological perspective) are excluded from this category.
- Label *v* (for *violence-related*): choose this label if the sentence depicts explicit sexual harassment, rape, lack of consent, or any other sort of non-consensual violence which concerns sexual intentions or activities.
  Please note that a sentence can include violence which is not sex-related. Such instances are excluded from this category. Consensual BDSM behaviors are likely to fall under the label *erotic*.
- Label *u* (for *socially unacceptable behaviors*): choose this label if the sentence describes sexual behaviors considered illegal, generally taboo, or violative of social norms, such as zoophilia, necrophilia, pedophilia, incest, and other sexual deviations.
  Please note that if you classify a sample as both *violence related* and featuring *socially unacceptable behaviors*, the priority should be given to the latter. Such instances should be labeled as *socially unacceptable behaviors*.
- Label *a* (for *ambiguous*): choose this label for sentences that may evoke some degree of erotic connotation but whose interpretation depends heavily on the context. If the sentence can reasonably be interpreted as non-erotic in a neutral context, it should be labeled as *ambiguous*.
- Label *n* (for *neutral*): choose this label if the sentence does not meet the criteria of the above mentioned sex-related categories. This typically includes non-sexual content on various topics, as well as general statements about human sexuality or intimate anatomy that do not carry erotic connotations.

C Prompt

Prompt Structure:

You are a classifier, and your task is to analyze the text and assign one of the following labels:

1. Neutral text
2. Erotic text
3. Ambiguous text
4. Socially unacceptable behavior text
5. Violence-related text

Based on the text below, return the results in JSON format, which will contain the keys `"Text"` and `"Prediction"`. The value in the `"Prediction"` field should contain only the name of the class.
Input text:

Figure 2: Classification prompt template used for evaluating language models (translated into English).

D Analysis of Type I and Type II errors

D.1 Full

Model	Erotic		Ambiguous		Violence		Unacceptable		Neutral	
	Type I	Type II	Type I	Type II	Type I	Type II	Type I	Type II	Type I	Type II
HerBERT Base	44	42	29	39	2	4	0	2	58	46
HerBERT Large	55	38	18	47	0	4	0	8	62	38
RoBERTa Base	35	42	41	40	3	2	2	1	50	46
RoBERTa Large	39	41	27	41	10	2	0	3	51	40
0-shot										
GPT-4o	97	26	14	72	32	3	228	7	1	266
C4AI Command-R	165	89	99	61	18	4	329	8	9	519
Llama 3.1 70B-Instruct	134	19	6	72	21	3	245	8	8	312
Mixtral 8x22B	91	73	12	72	43	2	244	5	14	252
Bielik-11B-v2.3-Instruct	260	10	6	72	29	2	144	7	0	348
PLLuM-Mistral-12B	55	108	34	70	11	1	50	8	104	64
PLLuM-Mixtral-8x7B	53	85	35	69	11	1	51	7	72	60
Llama-3.1-8B-PLLuM	5	238	1	72	2	3	7	8	312	6
PLLuM-Mistral-12B (SFT)	43	34	18	72	5	0	106	8	23	80
Llama-3.1-8B-PLLuM (SFT)	47	26	39	72	1	3	76	8	27	49
1-shot										
GPT-4o	99	26	14	72	33	3	255	7	1	296
C4AI Command-R	237	30	4	71	18	4	143	8	31	320
Llama 3.1 70B-Instruct	83	50	15	72	21	3	250	6	8	246
Mixtral 8x22B	87	70	19	71	26	2	178	6	30	185
Bielik-11B-v2.3-Instruct	189	14	14	71	19	3	104	6	4	236
PLLuM-Mistral-12B	52	132	50	70	7	3	54	8	119	65
PLLuM-Mixtral-8x7B	66	100	63	67	8	2	90	6	64	109
Llama-3.1-8B-PLLuM	53	216	20	71	36	3	44	7	260	114
PLLuM-Mistral-12B (SFT)	35	52	34	70	7	1	102	8	23	65
Llama-3.1-8B-PLLuM (SFT)	65	24	38	72	1	3	68	8	28	62
5-shot										
GPT-4o	90	34	26	72	43	1	231	8	1	276
C4AI Command-R	205	105	8	72	12	3	39	8	133	209
Llama 3.1 70B-Instruct	85	53	15	71	24	1	272	8	6	269
Mixtral 8x22B	142	48	16	72	33	2	154	5	17	233
Bielik-11B-v2.3-Instruct	172	19	7	72	24	2	138	6	1	239
PLLuM-Mistral-12B	55	108	34	70	11	1	50	8	104	64
PLLuM-Mixtral-8x7B	53	85	35	69	11	1	51	7	72	60
Llama-3.1-8B-PLLuM	30	288	16	72	20	3	10	8	277	51
PLLuM-Mistral-12B (SFT)	43	34	18	72	5	0	106	8	23	80
Llama-3.1-8B-PLLuM (SFT)	75	28	42	72	3	3	73	8	31	86

Table 8: Detailed Type I and Type II error counts for various models for *Full* dataset type.

D.2 Extended

Model	Erotic		Ambiguous		Violence + Unacceptable		Neutral	
	Type I	Type II	Type I	Type II	Type I	Type II	Type I	Type II
HerBERT Base	41	45	34	40	3	7	58	44
HerBERT Large	33	58	41	33	6	3	54	40
RoBERTa Base	45	49	38	42	8	2	48	51
RoBERTa Large	42	37	30	43	4	4	47	44
0-shot								
GPT-4o	95	33	240	34	42	10	2	262
C4AI Command-R	75	216	26	71	9	11	265	76
Llama 3.1 70B-Instruct	75	101	220	34	22	9	59	232
Mixtral 8x22B	52	99	225	36	39	9	39	174
Bielik-11B-v2.3-Instruct	159	42	66	63	19	10	44	160
PLLuM-Mistral-12B	78	91	40	56	100	5	55	104
PLLuM-Mixtral-8x7B	116	44	62	39	43	9	26	112
Llama-3.1-8B-PLLuM	7	256	3	72	2	12	334	6
PLLuM-Mistral-12B (SFT)	31	47	68	12	16	12	42	37
Llama-3.1-8B-PLLuM (SFT)	58	24	73	36	10	12	28	49
1-shot								
GPT-4o	117	28	241	37	44	10	1	293
C4AI Command-R	221	32	149	70	21	10	35	310
Llama 3.1 70B-Instruct	82	59	201	32	42	7	12	239
Mixtral 8x22B	87	66	166	43	80	7	27	185
Bielik-11B-v2.3-Instruct	194	27	95	64	35	9	6	213
PLLuM-Mistral-12B	56	105	39	65	56	5	93	55
PLLuM-Mixtral-8x7B	67	89	96	42	56	7	56	102
Llama-3.1-8B-PLLuM	60	208	65	71	32	11	255	120
PLLuM-Mistral-12B (SFT)	43	40	122	29	21	11	29	61
Llama-3.1-8B-PLLuM (SFT)	56	105	39	65	56	5	93	55
5-shot								
GPT-4o	88	44	256	35	62	9	2	281
C4AI Command-R	193	110	70	33	20	8	136	225
Llama 3.1 70B-Instruct	85	52	272	31	48	7	8	258
Mixtral 8x22B	130	110	70	33	20	8	136	225
Bielik-11B-v2.3-Instruct	153	23	163	53	27	9	2	239
PLLuM-Mistral-12B	64	101	50	42	38	10	100	68
PLLuM-Mitral-8x7B	66	89	66	41	44	10	76	76
Llama-3.1-8B-PLLuM	19	237	24	68	25	10	296	36
PLLuM-Mistral-12B (SFT)	60	28	87	34	26	11	19	80
Llama-3.1-8B-PLLuM (SFT)	72	26	70	44	22	10	27	80

Table 9: Detailed Type I and Type II error counts for various models for *Extended* dataset type.

D.3 Core

Model	Erotic		Ambiguous		Neutral	
	Type I	Type II	Type I	Type II	Type I	Type II
HerBERT Base	53	40	47	55	56	54
HerBERT Large	48	36	48	45	44	59
RoBERTa Base	37	46	46	46	62	51
RoBERTa Large	43	40	49	43	51	55
0-shot						
GPT-4o	74	25	301	23	1	279
C4AI Command-R	68	238	45	68	284	88
Llama 3.1 70B-Instruct	57	96	320	30	48	257
Mixtral 8x22B	58	90	241	30	35	172
Bielik-11B-v2.3-Instruct	142	21	109	59	20	178
PLLuM-Mistral-12B	87	39	98	52	36	110
PLLuM-Mixtral-8x7B	106	32	125	49	17	144
Llama-3.1-8B-PLLuM	10	239	8	72	305	12
PLLuM-Mistral-12B (SFT)	22	56	67	34	60	21
Llama-3.1-8B-PLLuM (SFT)	32	30	79	39	38	47
1-shot						
GPT-4o	95	21	327	28	0	329
C4AI Command-R	206	37	232	62	36	375
Llama 3.1 70B-Instruct	62	55	243	22	11	239
Mixtral 8x22B	79	61	229	31	18	193
Bielik-11B-v2.3-Instruct	140	19	181	52	2	232
PLLuM-Mistral-12B	89	57	86	60	67	113
PLLuM-Mixtral-8x7B	70	55	169	39	24	136
Llama-3.1-8B-PLLuM	72	197	92	69	232	127
PLLuM-Mistral-12B (SFT)	34	43	70	33	50	39
Llama-3.1-8B-PLLuM (SFT)	38	24	96	31	34	58
5-shot						
GPT-4o	88	29	303	29	0	290
C4AI Command-R	185	117	129	71	138	236
Llama 3.1 70B-Instruct	67	57	269	19	5	265
Mixtral 8x22B	105	37	253	41	16	243
Bielik-11B-v2.3-Instruct	106	31	240	44	1	244
PLLuM-Mistral-12B	102	54	62	62	67	97
PLLuM-Mixtral-8x7B	45	75	165	38	40	103
Llama-3.1-8B-PLLuM	24	247	43	69	290	38
PLLuM-Mistral-12B (SFT)	56	28	63	38	31	50
Llama-3.1-8B-PLLuM (SFT)	45	24	64	40	46	49

Table 10: Detailed Type I and Type II error counts for various models for *Core* dataset type.

D.4 Basic

Model	Erotic		Neutral	
	Type I	Type II	Type I	Type II
HerBERT Base	34	34	34	34
HerBERT Large	31	28	28	31
RoBERTa Base	31	23	23	31
RoBERTa Large	25	29	29	25
0-shot				
GPT-4o	115	1	1	115
C4AI Command-R	76	191	191	76
Llama 3.1 70B-Instruct	152	19	19	152
Mixtral 8x22B	95	33	33	95
Bielik-11B-v2.3-Instruct	140	18	18	140
PLLuM-Mistral-12B	87	20	20	87
PLLuM-Mixtral-8x7B	127	4	4	127
Llama-3.1-8B-PLLuM	6	246	246	6
PLLuM-Mistral-12B (SFT)	17	34	34	17
Llama-3.1-8B-PLLuM (SFT)	31	20	20	31
1-shot				
GPT-4o	132	1	1	132
C4AI Command-R	263	34	34	263
Llama 3.1 70B-Instruct	153	10	10	153
Mixtral 8x22B	134	18	18	134
Bielik-11B-v2.3-Instruct	149	1	1	149
PLLuM-Mistral-12B	127	35	35	127
PLLuM-Mixtral-8x7B	79	19	19	79
Llama-3.1-8B-PLLuM	76	156	156	76
PLLuM-Mistral-12B (SFT)	40	47	40	
Llama-3.1-8B-PLLuM (SFT)	127	35	35	127
5-shot				
GPT-4o	113	0	0	113
C4AI Command-R	199	96	96	199
Llama 3.1 70B-Instruct	159	4	4	159
Mixtral 8x22B	167	10	10	167
Bielik-11B-v2.3-Instruct	135	3	3	135
PLLuM-Mistral-12B	111	46	46	111
PLLuM-Mixtral-8x7B	44	32	32	44
Llama-3.1-8B-PLLuM	28	200	200	28
PLLuM-Mistral-12B (SFT)	73	15	15	73
Llama-3.1-8B-PLLuM (SFT)	43	17	17	43

Table 11: Detailed Type I and Type II error counts for various models for *Basic* dataset type.

E Computational Details

The details provided below correspond to the **Full** version of the dataset.

The encoder-based models were trained for approximately 5 hours and 10 hours, respectively, for the Base and Large versions, using A100 (40GB) GPU with early stopping. The batch size was set to 8 for both model variants. Training of the PLLuM family models on our dataset took approximately 3 hours for the Llama-3.1-8B-PLLuM (SFT) model and 4 hours for the PLLuM-Mistral-12B (SFT) model. These models were trained on a distributed setup consisting of 2 nodes, each equipped with 4 H100 (96GB) GPUs, ensuring parallel processing across GPUs for enhanced training efficiency.